Predicting Loan Payback

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1. Topic exploration: what each topic could do for a real organization

I considered all six topic areas. For each, one realistic example and the decision it could support:

Topic area	Realistic example	What the organization could learn or do
Supervised machine learning	**Banking.** A bank predicts whether a borrower will repay a loan, using income, debt-to-income ratio, credit score, and loan terms.	Rank borrowers by risk, set interest rates, and send borderline files to a loan officer.
Unsupervised machine learning	**Retail.** A supermarket groups loyalty-card customers by what, when, and how often they buy.	Give each group a different discount to increase sales, for example bulk coupons for family shoppers.
Dimensionality reduction	**Manufacturing.** A factory uses PCA to combine 40 related machine sensor readings into 2–3 summary scores.	Spot machines drifting away from normal and repair them before they break. Kept: the overall pattern. Lost: which single sensor caused it.
Natural language processing	**Telecom.** A phone company sorts customer support messages into billing, network outage, or cancellation.	Send each message to the right team and track which problems grow each week.
Semantic or sentiment analysis	**Hospitality.** A hotel chain scores guest reviews as positive or negative for room, staff, and breakfast.	Find the service that upsets guests most and fix it first.
Social media / digital channel analytics	**Consumer brand.** A coffee chain compares the likes, shares, and reach of its own Instagram posts.	Choose which posts (new drinks, promotions) and posting times to use next.

Credit scoring: from expert judgment to explainable machine learning

Supervised classification has been used to predict loan repayment for nearly 70 years. What changed is the model, and what never changed is the duty to explain.

Topic: supervised ML
Industry: consumer lending

ORIGINS AND DEVELOPMENT

Scoring industry · Machine learning · Regulation

Late 1950s
Credit scoring first emerges to support credit decisions at large retail stores and finance companies; by the end of the 1970s, most of the largest US commercial banks use scoring systems [1].

1974
The Equal Credit Opportunity Act (ECOA). Under Regulation B, reasons given for an adverse action "must be specific and indicate the principal reason(s)" [2].

Late 1980s
Generic credit scores based only on credit-bureau records appear and work across many loan products [1].

2004
Basel II lets banks use their own internal rating models to estimate the probability of default for capital rules [3].

2016–2017
XGBoost and LightGBM make gradient-boosted trees fast and accurate on large tables [4][5]. SHAP explains each prediction feature by feature [6].

2024
U.S. Treasury: AI "has long been deployed" in credit underwriting [10].

PRACTICAL USE TODAY

52% of surveyed banks use ML in production for credit risk [8]

25% more are piloting ML for credit risk [8]

43 banks in 9 regions surveyed, including the U.S. (2022) [8]

WHO USES IT, AND FOR WHAT

- **Lenders:** use machine learning in credit underwriting, including to analyze alternative data such as rent and utility payments [10].
- Explainability is a primary challenge for ML in credit risk [8].

VALUE AND IMPACT

POSSIBLE BENEFITS

In a 2025 study, a machine learning model using credit-bureau and cash-flow data was the most predictive and approved the most applicants, while keeping approvals of borrowers who later defaulted low [7].

COSTS AND RISKS

Black-box models give limited insight, and poorly trained AI can reinforce bias [10]; complex models are harder to validate.

Benefits depend on the lender and data; they are not guaranteed savings.

CONNECTION TO MY PROJECT

- **Champion vs. challenger:** the data includes the lender's own grade (A1 to F5), so I can test ML against the lender's current ranking, like a real model review.
- **Method:** logistic regression (transparent) and LightGBM (gradient boosting [5]).
- **Explainable:** logistic regression coefficients and LightGBM feature importance show which factors drive default risk.
- **Early evidence:** AUC 0.922 vs. 0.655; about $1.2B more lending at the same default losses (synthetic data; Section 5).

[1] Federal Reserve Board (2007), Report to the Congress on credit scoring.
[2] CFPB, Regulation B, 12 CFR §1002.9.
[3] Basel Committee on Banking Supervision (2004), Basel II revised framework.
[4] Chen & Guestrin (2016), XGBoost, KDD.
[5] Ke et al. (2017), LightGBM, NeurIPS.
[6] Lundberg & Lee (2017), SHAP, NeurIPS.
[7] FinRegLab (2025), Machine learning & cash flow data in consumer underwriting.
[8] IIF & EY (2022), Survey on machine learning in credit risk.
[10] U.S. Treasury (Dec. 2024), AI in financial services report.
Full links in the reference list at the end of this document.

3. Selected focus and project proposal

> **Business problem.** Imagine a consumer lender with **$8.9 billion** in approved personal loans. One in five loans defaults, leaving **$1.8 billion** in unpaid principal. The lender's risk grade cannot separate safe borrowers from risky ones, so it approves too many bad loans and turns away good ones. A better risk ranking could add about **$1.2 billion** in lending at the same default losses and earn about **$161 million** in additional interest income.

> **Question.** Can a machine learning model rank borrowers by default risk better than the lender's own grade, so the lender can lend more without higher default losses, while keeping decisions explainable?

Element	My project
Stakeholder	A consumer lender's credit risk manager (sets policy) and loan officers (review borderline files).
Label	`loan_paid_back`: 1 = repaid, 0 = not repaid. Binary classification. Positive class = **not repaid** (the outcome we want to catch).
Intended use	Route similar future applicants: **approve / review / reject**, with cutoffs set by business costs; show loan officers the main risk drivers.

Success measures

Measure	Useful if
ROC AUC (ranking)	Above the lender's grade (0.655) and a classic credit score + DTI scorecard (0.768).
Recall on loans not repaid	Of the loans that default, how many are caught? Higher than every baseline.
Dollar impact	More lending and interest income at the same defaulted principal; less defaulted principal at the same lending volume.
Interpretability	The main risk drivers are clear and agree with the EDA.

Dataset and access

- **Source and access:** Kaggle Playground Series S5E11, "Predicting Loan Payback" [9]. Sign in to Kaggle, accept the competition rules, and download the files into `data/` (steps in the README). The rules do not allow redistributing the data, so the public repository holds code, figures, and results only.
- **One row** = one loan that was already approved: the borrower's finances, loan terms, profile, and whether it was paid back.

Key field	Business meaning	Expected signal
employment_status	Stability of income	Unemployed or student → much higher risk
debt_to_income_ratio	Existing debt payments ÷ income	Higher DTI → higher risk
credit_score	Bureau score (395 to 849)	Higher score → lower risk
grade_subgrade	Lender's own grade, A1 (best) to F5 (worst)	The lender's current ranking
annual_income, loan_amount, interest_rate	Ability to repay, exposure, price of risk	Numeric features
marital_status, education_level, loan_purpose	Profile and use of funds	Weak signals
gender	Borrower gender	Excluded; fairness check only

Limits. (1) **Synthetic:** generated from an original loan dataset, so dollars show direction, not a real bank's results. (2) **Approved loans only:** declined applicants are not observed. (3) **Privacy:** no personal identifiers. (4) **Licensing:** Kaggle rules prohibit sharing the data with non-participants. (5) **Dollar assumptions:** 60% loss given default; one year of interest income; portfolio figures scale the 20% validation set by 5.

Planned method

Step	What I do
Shortlist	For tabular data: a logistic model plus a tree ensemble.
Baselines	(1) Majority baseline: always predict "repaid"; (2) the lender's grade only; (3) a classic credit score + DTI scorecard.
Logistic regression	A transparent first challenger: $z = b + w \cdot x$, $p = 1 / (1 + e^{-z})$, trained with log loss (maximum likelihood). Coefficients show the direction and size of each risk factor.
Gradient boosting	LightGBM adds trees sequentially; a strong tabular model that can learn the nonlinear DTI threshold.
Preprocessing	Learn the preparation once from training rows and reuse it: ColumnTransformer → Pipeline → cross-validation. The scaler and encoder are fit only on the training partition to prevent leakage.
Evaluation	Fit on training data, choose on validation, report on untouched test data. **Model comparison:** stratified 5-fold cross-validation on the training data. **Dollar impact and cutoffs:** a 20% validation set held out from the training data. **Kaggle test set:** has no labels, so it is only used for the final Kaggle submission.

4. Initial data exploration

Check	Result	Decision
Size	593,994 training rows × 13 columns; 254,569 test rows	Enough for a fixed 20% validation set
Missing / duplicates	None	No imputation needed
Target balance	79.9% repaid, 20.1% not repaid	Accuracy is misleading; use ROC AUC and recall
Categories	No spelling or case clashes; same levels in train and test	Keep "Other" as a real category
Outliers (Tukey 1.5 × IQR)	About 3% of income and DTI values	Keep: high-DTI outliers repaid 45% vs. 81%; they are real high-risk borrowers
Skewness	Income and DTI right-skewed	Log transform
Correlation	Credit score vs. grade: r = −0.91	Logistic regression keeps credit score only

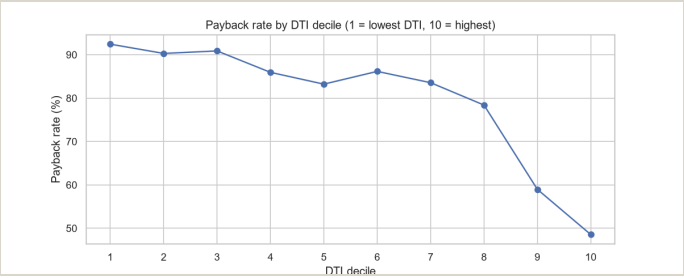


Figure 1. Payback rate by DTI decile.

Interpretation. Payback is 83–92% in the first seven deciles, then falls to 78%, 59%, and 49%. Risk is nonlinear, so a tree model is a good fit.

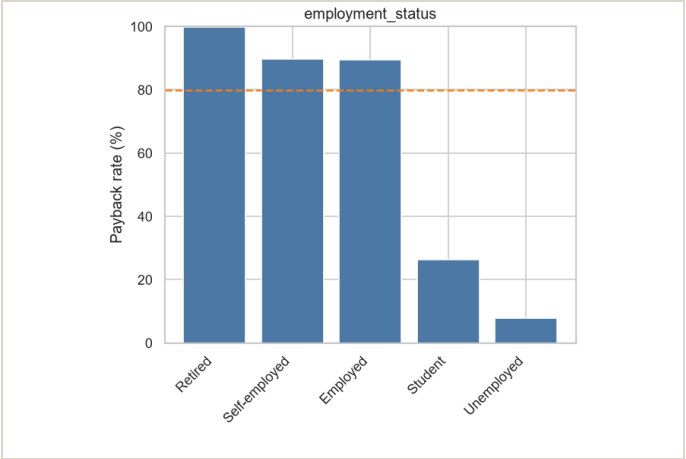


Figure 2. Payback rate by employment status (dashed line = 79.9%).

Interpretation. The strongest signal: 7.8% of unemployed borrowers repaid vs. 89.4% of employed. Other categories barely change the payback rate.

Full audit: `01_eda.ipynb`, `src/eda_audit.py`, and `reports/`.

5. Baseline and proof of concept

Stratified 5-fold cross-validation on the training data (`02_modeling.ipynb`):

Model	ROC AUC	Recall (not repaid)
Majority baseline: always "repaid"	0.500	0.0%
Lender's grade only	0.655	1.8%
Classic scorecard: credit score + DTI	0.768	19.3%
Logistic regression	0.908	59.8%
LightGBM	**0.922**	**60.9%**

Recall at the default 0.5 threshold. The majority baseline is 79.9% accurate but catches no defaults.

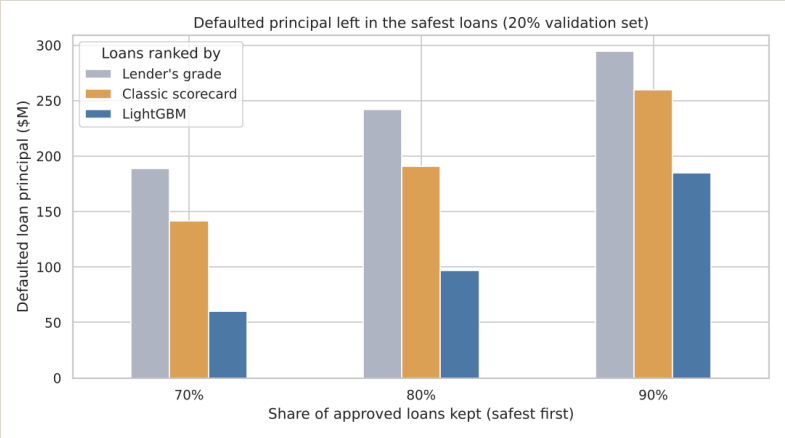


Figure 3. Defaulted principal left in the safest loans (20% validation set, 118,799 loans).

Cut losses. Keeping the safest 80% of approved loans, LightGBM leaves **$97.1M** of defaulted principal vs. **$242.0M** with the lender's grade: about **60% less**. The gap holds at 70% and 90%.

Synthetic data: the direction matters more than the exact dollars.

Grow revenue: same defaulted principal, more lending

Ranked by	Loans approved	Lending	Defaulted principal	Interest (1 year)
Lender's grade	80.0%	$1,426.7M	$242.0M	$141.3M
LightGBM	**93.4%**	**$1,663.2M**	**$242.0M**	**$173.6M**

At the same defaulted principal, LightGBM approves **15,880 more loans**: **$236M more lending** and **$32.3M more interest** on the validation set. Scaled to the whole portfolio, that is about **$1.2B more lending** and **$161M more interest**.

First result	Illustrative
Risk drivers	Both models agree on the top three: employment status (68.5% of LightGBM importance; unemployed and students repay far less), debt-to-income ratio (19.3%), and credit score (8.6%).
Approve / review / reject	Two cutoffs on the payback probability, chosen with assumed business costs: approve ≥ 0.89, review 0.77–0.89, reject < 0.77. Default rates on the validation set: 3.7% / 15.4% / 67.1%. Preliminary: chosen and measured on the same validation set.

Feasibility. The models run in minutes and already meet the AUC and dollar-impact success measures.

6. Open questions and next steps

Still unknown

- **Calibration:** a strong ROC AUC does not prove that a predicted 80% means 80% in reality.
- **Fairness:** do errors differ across gender groups?
- **Real data:** does a model trained on synthetic rows hold up on the original loan dataset?

Next steps

Step	What I will do
Confusion matrix	Name each kind of mistake at the chosen cutoffs; report precision and recall with counts.
Thresholds	Choose approve / review / reject cutoffs with business costs on the validation set, then confirm them on a separate held-out set.
Calibration and fairness	Calibration curve; compare errors across gender groups.
Tuning	Tune LightGBM with cross-validation only.
Model handoff	Save the fitted pipeline (joblib) with its preprocessing and metadata; reload it and reproduce a reference prediction.
Local API	FastAPI `/predict`: one applicant in; payback probability, risk tier, and top drivers out; invalid input returns 422; packaged in Docker.
Communicate	Data-to-decision pipeline visual, written report, and stakeholder presentation.

Key terms

ROC AUC: How well the model ranks repaying borrowers above defaulting ones, from 0.5 (random guess) to 1.0 (perfect).

Accuracy: The share of all predictions that are correct, which is misleading when most loans are repaid.

Recall (not repaid): The share of defaulted loans that the model correctly flags.

Majority baseline: A model that always predicts the most common outcome, "repaid."

Lender's grade: The lender's own risk rating, from A1 (safest) to F5 (riskiest).

Scorecard: A simple credit model that adds up points from a few factors, such as credit score and DTI.

DTI: Debt-to-income ratio: a borrower's debt payments divided by income.

Defaulted principal: The loan amount that was not repaid.

Loss given default: The share of a defaulted loan the lender actually loses after recovery.

Logistic regression: A model that turns a weighted sum of factors into a probability between 0 and 1.

Gradient boosting / LightGBM: A model that adds many small decision trees one by one, each fixing earlier errors.

Cross-validation (5-fold): Training and testing the model five times on different splits so every row is tested once.

Validation set: Data held back from training to choose settings such as cutoffs.

Data leakage: Test information slipping into training, which makes results look better than they are.

Calibration: Whether predicted probabilities match reality, for example whether 80% of borrowers scored 80% actually repay.

Feature importance: How much each factor helps the model separate repaid from defaulted loans.

PCA: Principal component analysis: a method that combines many related variables into a few summary scores.

References

[1] Board of Governors of the Federal Reserve System. (2007, August). *Report to the Congress on credit scoring and its effects on the availability and affordability of credit*. https://www.federalreserve.gov/boarddocs/RptCongress/creditscore/general.htm

[2] Consumer Financial Protection Bureau. (n.d.). Regulation B (Equal Credit Opportunity Act), 12 CFR §1002.9: Notifications. https://www.consumerfinance.gov/rules-policy/regulations/1002/9/

[3] Basel Committee on Banking Supervision. (2004, June). Basel II: International convergence of capital measurement and capital standards: A revised framework. Bank for International Settlements. https://www.bis.org/publ/bcbs107.htm

[4] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD Conference*. https://arxiv.org/abs/1603.02754

[5] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems 30*. https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf

[6] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems 30*. https://arxiv.org/abs/1705.07874

[7] FinRegLab. (2025). *Advancing the credit ecosystem: Machine learning & cash flow data in consumer underwriting*. https://finreglab.org/research/advancing-the-credit-ecosystem-machine-learning-cash-flow-data-in-consumer-underwriting/

[8] Institute of International Finance & EY. (2022, December). *IIF and EY survey report on machine learning: Uses in credit risk and AML applications* (public summary). https://www.iif.com/Publications/ID/5184/IIF-and-EY-2022-Survey-Report-on-Machine-Learning-Summary

[9] Kaggle. (2025). Predicting Loan Payback: Playground Series, Season 5, Episode 11. https://www.kaggle.com/competitions/playground-series-s5e11

[10] U.S. Department of the Treasury. (2024, December). *Artificial intelligence in financial services*. https://home.treasury.gov/system/files/136/Artificial-Intelligence-in-Financial-Services.pdf